

$$c(x) = (\cos 4\theta(x), \sin 4\theta(x))$$

- **4-RoSy field:** invariant under 90° rotation
- **Representative vector** encodes direction modulo $\pi/2$
- **Boundary-aligned** via Dirichlet conditions

Literature: Knöppel et al., “Globally optimal direction fields,” SIGGRAPH 2013

Code: `scripts/stage1/partition_surface.py` — cross-field solver pipeline